NATIONAL INSTITUTE OF TECHNOLOGY, TIRUCHIRAPPALLI

I B.Tech. (ICE – B Section)

PHYSICS (PHIR11) – I Cycle Test – July 2026 Session

Date: 12.10.2026 *Time: 4.30 – 5.30 pm* *Marks: 20*

1 Write down the Schrodinger equation for a particle of mass m confined in a potential $V(x)$. If $\psi(x,t)$ is the solution describing quantum state of the particle at time t, define expectation values $\langle x\rangle$ and $\langle p\rangle$ of the particle and show that $\langle p\rangle = m\frac{d}{dt}\langle x\rangle$. What do you infer from the relation? (4)

2 Find the momentum expectation value $\langle p\rangle$ of a particle in its n$^{\text{th}}$ eigenstate, if it is confined in one-dimensional box of length L. What do you infer from the result? (4)

3 A measurement establishes the position of a proton with an accuracy of $\pm 2\times 10^{-11}$ m. Find the uncertainty in the position of proton after 2 sec. Assume that the velocity of proton $v \ll c$. (4)

4 Construct a wave packet by adding two sinusoidal travelling waves with small difference in their wavenumbers and frequencies. Define phase and group velocities of the wave packet. Show that such a wave packet can represent a particle moving with velocity v. (4)

5 Define binding energy of a nucleus. Show the variation of binding energy per nucleon with atomic mass. What do you understand from this? (4)

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